M339G: 03/27/2026.

Bivariate Normal Random Pairs.

(Pitman's "Probability")

Recall: In 1D, the standard normal density is

$$\boxed{\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad \text{for all } z \in \mathbb{R}}$$

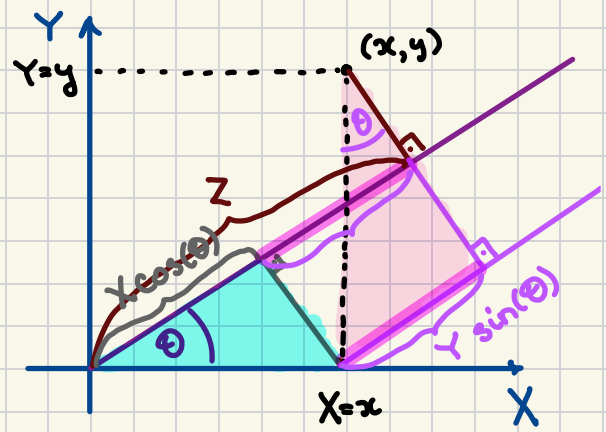
In 2D, we start w/ X and Y that are $N(0,1)$ and INDEPENDENT.

Their joint density is

$$f(x,y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)} \quad \text{for all } (x,y) \in \mathbb{R}^2.$$

Standard.

Start w/ a pair of independent, standard normal vs. Say, X and Y.



$\Rightarrow$ $\boxed{Z = X \cdot \cos(\theta) + Y \cdot \sin(\theta)}$

$\Rightarrow$ $Z \sim \underline{N(\cdot,\cdot)}$

$\mathbb{E}[Z] = 0,$

$Var[Z] = 1.$

$\boxed{Z \sim N(0,1)}$

Q: What is the correlation coefficient between X and Z?

$\rightarrow$:

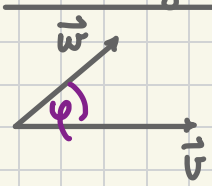
$$\rho_{X,Z} = \frac{Cov[X,Z]}{\underbrace{SD[X]}_{1} \cdot \underbrace{SD[Z]}_{1}} = Cov[X,Z]$$

$$\rho_{X,Z} = \mathbb{E}\left[(X-\mu_X)\cdot(Z-\mu_Z)\right]$$
$$= \mathbb{E}[X\cdot Z] - \underbrace{\mu_X\cdot\mu_Z}_{=0}$$
$$= \mathbb{E}[X\cdot Z]$$
$$= \mathbb{E}\left[X\cdot(X\cdot\cos(\Theta) + Y\cdot\sin(\Theta))\right] =$$
$$= \cos(\Theta)\cdot\underbrace{\mathbb{E}[X^2]}_{=1} + \sin(\Theta)\cdot\underbrace{\mathbb{E}[X\cdot Y]}_{=0}$$

$$\boxed{\rho_{X,Z} = \cos(\Theta)}$$

$\square$

Linear Algebra



$$\vec{v}\cdot\vec{w} = \|\vec{v}\|\cdot\|\vec{w}\|\cdot\cos(\varphi)$$

$$\boxed{\cos(\varphi) = \frac{\vec{v}\cdot\vec{w}}{\|\vec{v}\|\,\|\vec{w}\|}}$$

Special Cases:

$\Theta = 0 \Rightarrow Z = X$

$\Theta = \frac{\pi}{2} \Rightarrow Z = Y$

so, X and Z are independent.

$\Theta = \pi \Rightarrow Z = -X$

In general: For each correlation coefficient

$$-1 \leq \rho \leq 1,$$

there exists an angle $\boxed{\Theta = \arccos(\rho)}$ such that X and Z defined as above have the correlation coefficient ρ.

Alternative:

$$Z = \underset{\cos(\theta)}{\rho} \cdot X + \underset{\sin(\theta)}{\sqrt{1-\rho^2}} \cdot Y \quad \longleftarrow \checkmark$$

w/ X and Y independent and $N(0,1)$.

Joint Density:

$$f_{X,Z}(x,z) = \frac{1}{2\pi(1-\rho^2)} \exp\left(-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xz + z^2)\right)$$

Marginal Dist'ns: $X \sim N(0,1)$, $Y \sim N(0,1)$

Conditional Dist'ns: Given $\boxed{X=x}$, $Z \sim \text{Normal}(\rho \cdot x, \text{var} = 1-\rho^2)$

Given $\boxed{Z=z}$, $X \sim \text{Normal}(\rho \cdot z, \text{var} = 1-\rho^2)$

Independence:

$$\boxed{\begin{array}{c} X \text{ and } Z \text{ independent} \\ \text{iff} \\ f_{X,Z} = 0 \end{array}}$$

Any Bivariate Normal.

A random pair (U,V) is said to be bivariate normal w/ parameters $\mu_U, \mu_V, \sigma_U, \sigma_V$ and ρ if

$$\boxed{(X,Z) := \left(\frac{U-\mu_U}{\sigma_U}, \frac{V-\mu_V}{\sigma_V}\right)}$$

has the standard normal w/ correlation ρ.

Note:

$\rho_{U,V} = ?$

$$\boxed{\begin{array}{c}\text{Hope:}\\ \rho\end{array}}$$

$$\rho_{U,V} = \frac{Cov[U,V]}{SD[U]\cdot SD[V]} = \left\{\begin{array}{l} U = \mu_U + \sigma_U \cdot X \\ V = \mu_V + \sigma_V \cdot Z \end{array}\right\}$$

$$= \frac{Cov[\cancel{\mu_U} + \sigma_U \cdot X, \cancel{\mu_V} + \sigma_V \cdot Z]}{SD[\cancel{\mu_U} + \sigma_U \cdot X]\cdot SD[\cancel{\mu_V} + \sigma_V \cdot Z]}$$

$$= \frac{Cov[\sigma_U \cdot X, \sigma_V \cdot Z]}{SD[\sigma_U \cdot X]\cdot SD[\sigma_V \cdot Z]}$$

$$= \frac{\cancel{\sigma_U}\cdot\cancel{\sigma_V}\cdot Cov[X,Z]}{\cancel{\sigma_U}\cdot SD[X]\cdot\cancel{\sigma_V}\cdot SD[Z]} = \rho_{X,Z} = \rho$$

So, in particular,

$$\boxed{\begin{array}{c} U \text{ and } V \text{ are independent} \\ \text{iff} \\ \rho = 0 \end{array}}$$